

Dry-run mindset

Before a clean deletes build junk, preview what would change

Preview, read, then live

- Prefer preview first
- Read every path on the list
- Live run only after the list matches what you meant to remove
- Echo-first is the same idea, print the command before you paste the real one

Browser lab

Digital Design and Verification Platform

HomeToolsDry-run

Preview before you destroy --dry-run, -n, and echo-first habits for Make, rm, and git clean.

Starter example: Chip tree with junk in build/, logs/, and an untracked untracked.o. Prefer dry-run, preview cleans, then run live only when the list looks right.

Load starter example

Challenges 0 / 22 cleared

Quiz: Make preview: Make flag to print recipes without running? Answer: -n

Answer

Show hintCheckNextSolve

Quiz: Make preview

Quiz: git clean preview

Quiz: echo-first

Quiz: order

Preview dry on

make -n clean

rm dry

git clean -nd

Echo-first

Live after dry

Force live

git clean live

Quiz: rcyn

Success preview

Quiz: rmdest

Preview lists paths

Preview off is live

Quiz: live git clean

echo then rm

Make targets junk

Quiz: habit

Full habit

Care ideas

1. Preview2. Read the list3. Live only if OK

-n / -dry-run

Show what would run or delete -- no real change.

Echo-first

Print the command (echo rm ...) before pasting the real one.

Workspace

prefer dry: on • junk left: 5 • sources: ok

```
cli/alu.v # keep
ctrl_alu.v # keep
ctrl_alu.c # keep
makefile # keep
README.md # keep
build/ctrl_alu.o.v # junk
build/mem_vcd # junk
logs/alm.log # junk
new/memctrl.txt # junk
untracked.o # junk
```

Commands

☒ Prefer dry-run / n

(preview appears when you run a command)

make clean (or make -n clean)

rm -rf build/ logs/ (or dry)

git clean -fd (or -nd)

echo-first: echo rm -rf build/

Force live (ignore prefer)

Output

Alternative: junk or build/ logs/ mem.o
untracked.o - prefer dry-run

Cheat sheet

Tool	Preview flag
Make	make -n -clean -- print recipes, do not run
rm (GNU)	rm -n / prefer listing paths first
git clean	git clean -nd -- show, do not delete
rcynrc	rcynrc --dry-run ..
Any shell	echo the command before running it

- Dry-run never replaces reading the path list -- globs expand wider than you think.
- Keep prefer-dry-on until the preview matches only junk.
- Source under [git1/](#) and [lib/](#) should survive every clean here.

Dry-run lab

learn_unix — dry run lab

Real shell practice

```
wsl - module25-dry-run-lab/examples/dry_run

# Track A - real Unix
# real shell session (Track A)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module25-dry-run-lab/examples/dry_run

$ ls -la
total 4
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 00:49 ..
-rwxrwxrwx 1 yongfu yongfu 1508 Jul 20 09:15 README.md
-rwxrwxrwx 1 yongfu yongfu 124 Jul 20 09:15 commands_to_try.txt
-rwxrwxrwx 1 yongfu yongfu 688 Jul 18 16:34 remove_old.sh

$ chmod u+x remove_old.sh

$ ./remove_old.sh --help
Usage: ./remove_old.sh [--dry-run]

$ ./remove_old.sh --dry-run
[dry-run] Would remove 0 file(s). Run without --dry-run to remove.

$ touch -d 2020-01-01 old_file.txt

$ ls -la old_file.txt
-rwxrwxrwx 1 yongfu yongfu 0 Jan  1 2020 old_file.txt

$ ./remove_old.sh --dry-run
Would remove: ./old_file.txt
[dry-run] Would remove 1 file(s). Run without --dry-run to remove.

$ ./remove_old.sh
Removed 1 file(s).

$ ls -la
total 4
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 09:15 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 00:49 ..
-rwxrwxrwx 1 yongfu yongfu 1508 Jul 20 09:15 README.md
-rwxrwxrwx 1 yongfu yongfu 124 Jul 20 09:15 commands_to_try.txt
-rwxrwxrwx 1 yongfu yongfu 688 Jul 18 16:34 remove_old.sh
```

Real shell — dry-run remove

Real shell practice — try these

```
# pwd — print working directory (where am I?)  
pwd
```

```
# ls -la — list all entries, long format (what is here?)  
ls -la
```

```
# chmod u+x remove_old.sh — make the cleanup script executable  
chmod u+x remove_old.sh
```

```
# ./remove_old.sh --help — show usage (dry-run flag)  
./remove_old.sh --help
```

```
# ./remove_old.sh --dry-run — preview only; no deletes yet  
./remove_old.sh --dry-run
```

```
# touch -d 2020-01-01 old_file.txt — create a file older than 365 days
```

Pitfalls to watch

- Do not run a live clean because dry-run felt slow
- Do not ignore paths you do not recognize on the preview list
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, clear a few challenges after the starter
- On the real shell, dry-run the remove script, then live-run only when the list looks right
- When you are ready, take the short quiz, then continue to log and failure triage